DO NOW — RECALL

Lesson 091 · Fetch–Decode–Execute: A Register Trace

Name: _____ Class: _____ Date: _____

1. Which register holds the current instruction?

2. What is the role of MAR?

3. Where is the next instruction address kept?

4. A full adder has A=1,B=0 and carry-in=1. State Sum and carry-out.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

CIR.

2. Model answer

Stores address being accessed.

3. Model answer

PC.

4. Model answer

$1+0+1=10_2$, so Sum0, carry-out1.

By the end, you can…

• Fetch copies the address from PC to MAR; memory read transfers the instruction through MDR into CIR. In a sequential fetch, PC advances.

• The Control Unit decodes the current instruction held in CIR and produces the necessary control signals.

• Execution of LDA 80 reads the data held at memory address 80 into ACC; the next instruction is not executed until its own fetch.

Vocabulary — know these terms

fetch — Retrieve next instruction.

decode — CU interprets instruction.

execute — Carry out operation.

CIR — Current Instruction Register.

ACC — Accumulator.

NEW LEARNING

Read, annotate and explain

Observe

Fetch copies the address from PC to MAR; memory read transfers the instruction through MDR into CIR. In a sequential fetch, PC advances.

Explain

The Control Unit decodes the current instruction held in CIR and produces the necessary control signals.

Evaluate

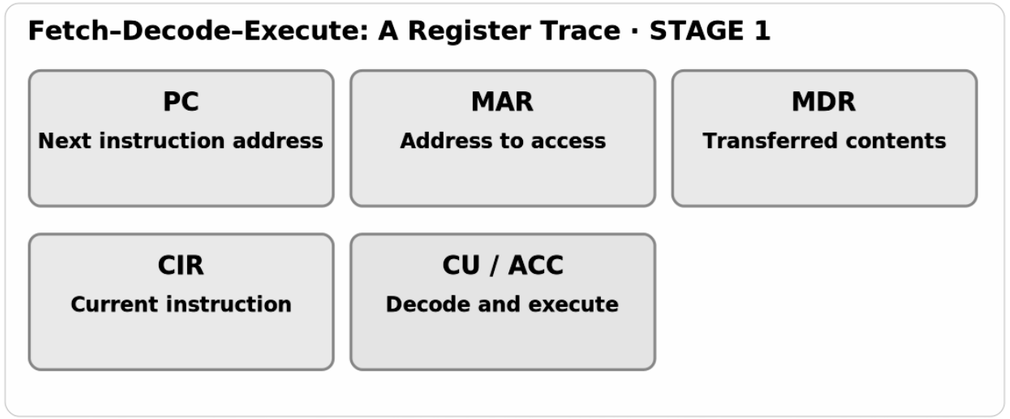
Execution of LDA 80 reads the data held at memory address 80 into ACC; the next instruction is not executed until its own fetch.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

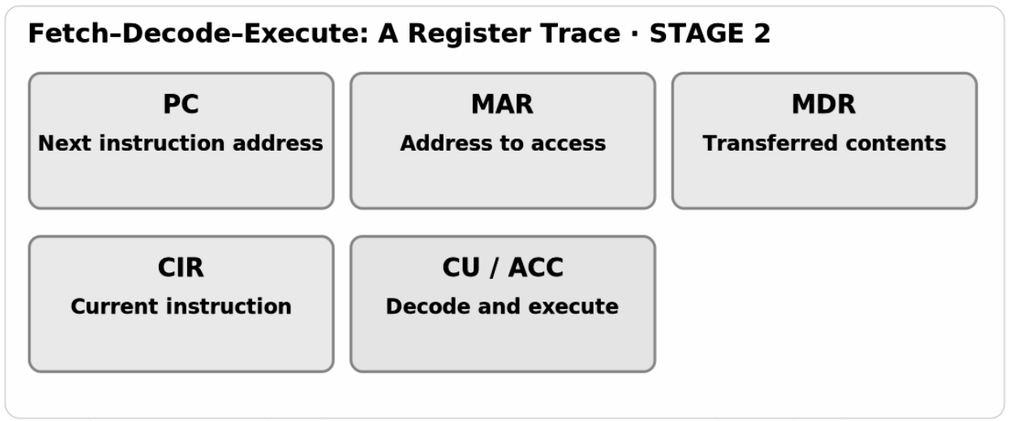
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Fetch–Decode–Execute: A Register Trace · STAGE 1; PC20 → MAR20 → MDR instruction LDA80 → CIR; PC21

New Learning



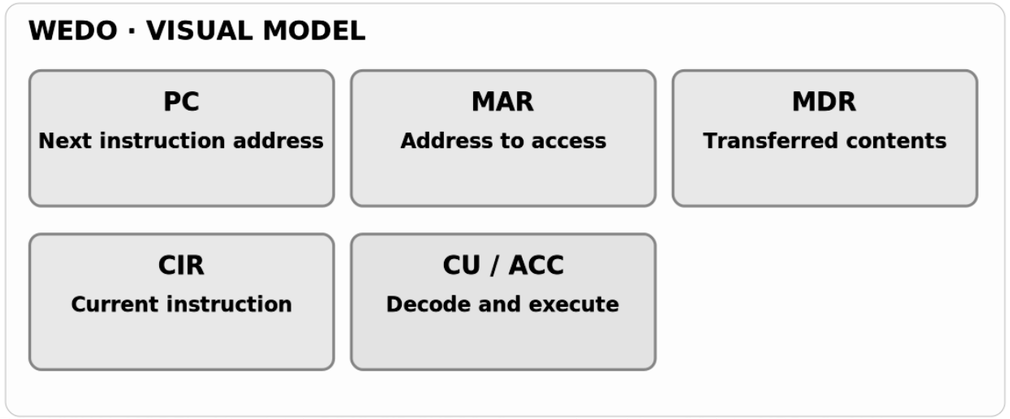
Fetch–Decode–Execute: A Register Trace · STAGE 2; PC20 → MAR20 → MDR instruction LDA80 → CIR; PC21; CU interprets LDA 80; identifies operand location

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

PC=42, memory[42]=SUB 90, memory[90]=3, ACC=11. Trace registers and result.

After fetch PC43,CIR SUB90. Operand read from90 into MDR=3; ALU computes11−3, ACC8.



WEDO · VISUAL MODEL; After fetch PC43,CIR SUB90. Operand read from90 into MDR=3; ALU computes11−3, ACC8.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

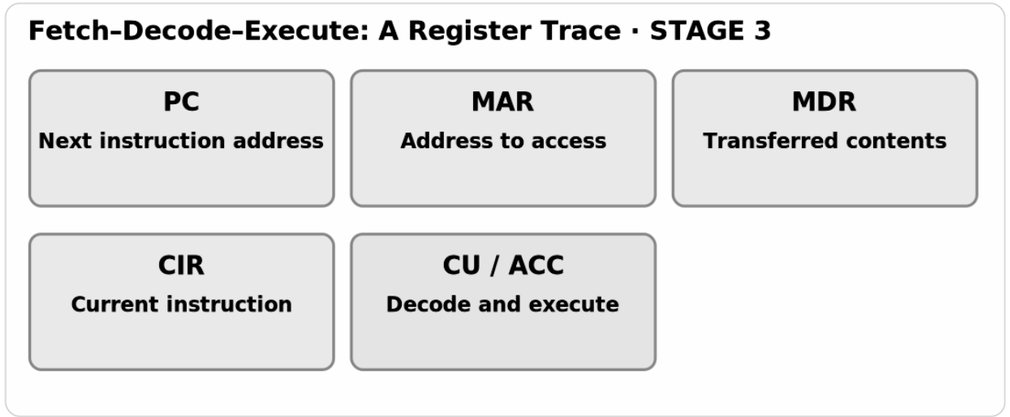
Guided 1 — CHECK

Explain the mechanism
After fetch PC43,CIR SUB90. Operand read from90 into MDR=3; ALU computes11−3, ACC8.
The addressed operand must not be confused with instruction address42.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Fetch–Decode–Execute: A Register Trace · STAGE 3; PC20 → MAR20 → MDR instruction LDA80 → CIR; PC21; CU interprets LDA 80; identifies operand location; MAR80 → MDR13 → ACC13; next instruction PC21

I DO — WORKED EXAMPLE

Follow the worked method

PC=30, memory[30]=ADD 70, memory[70]=5, ACC=4. Trace after fetch and execute.

Explain the mechanism

Fetch: MAR30, CIR ADD70, PC31. Execution reads memory[70]=5 and ALU adds to ACC4, giving ACC9.

The PC increments in fetch while the ALU operates on ACC and fetched operand.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

During a sequential fetch, MDR stores the instruction transferred from memory rather than its address.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

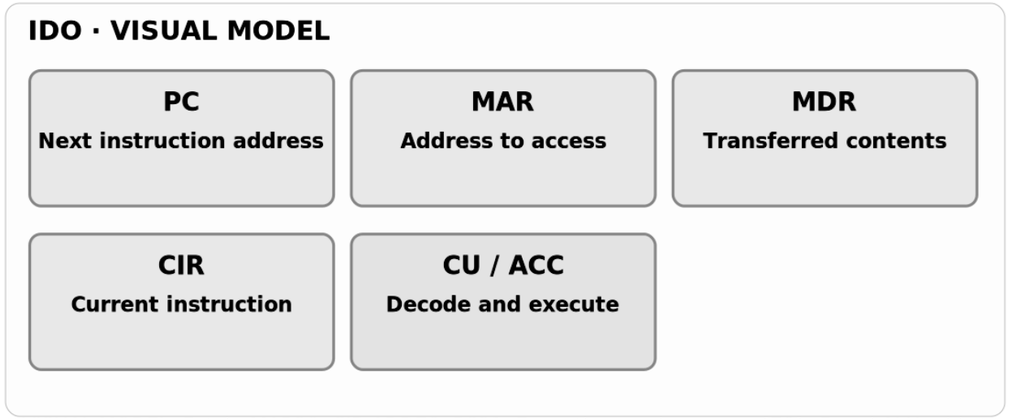
An answer states 'MDR stores address 80 and MAR stores the fetched 13'. Correct the error with a precise memory-read explanation. [6]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

PC=30, memory[30]=ADD 70, memory[70]=5, ACC=4. Trace after fetch and execute.

Fetch: MAR30, CIR ADD70, PC31. Execution reads memory[70]=5 and ALU adds to ACC4, giving ACC9.



IDO · VISUAL MODEL; Fetch: MAR30, CIR ADD70, PC31. Execution reads memory[70]=5 and ALU adds to ACC4, giving ACC9.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

PC=42, memory[42]=SUB 90, memory[90]=3, ACC=11. Trace registers and result.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Trace the fetch of instruction at PC20 into MAR,MDR,CIR; give PC after sequential increment. [6]

Question 2

Explain difference between decode and execute for LDA 80 when memory[80]=13. [6]